

Semester Test 2

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Electricity and Electronics EBN 122

20 October 2022

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	90 minutes	Open/Closed book:	Closed
Additional time allocations:	5 min before and 10 min after for OCR answer sheet completion	Allowable materials:	Pencil case, non-programmable calculator
Extra-time venue:	Eng III-6	Submission format:	OCR answer sheets 00/01 and 02/03
Total number of pages (this page included)		15	

Lecturers: Prof P. de Villiers, Prof P. Jacobs**Important**

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of OCR answer sheet 00/01.
4. Read the instructions on OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. Final answers must be written as real numbers and not as fractions.
7. Use **at least 3 significant digits** to write real numbers.

INSTRUCTIONS**Do step 0 in the first 5 minutes of the session. (-5 to 0 min)****Do steps 1 and 2 in the next 100 minutes of the test. (0-100 min)****TIME: -5 min to 0 min****STEP 0 :** Complete the front page of OCR answer sheet 01, and also fill in your student number on additional OCR answer sheets.**TIME: 0 min to 100 min****STEP 1:** Do your calculations on the question paper.**STEP 2:** Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

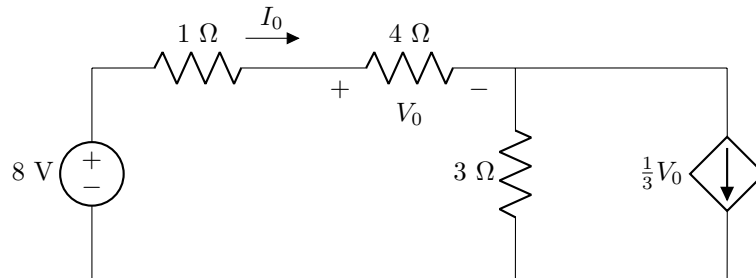
- The Assessment ID is: **2022EBN122S02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

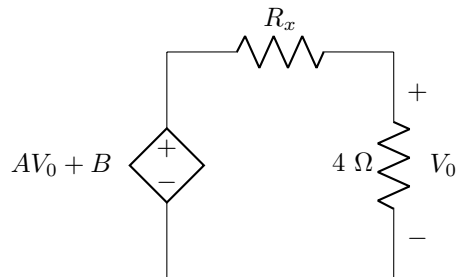
Questions 1 to 3 [4]

Use source transformation to obtain Circuit B from Circuit A. Determine R_x , and the coefficients A and B .
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Circuit A:



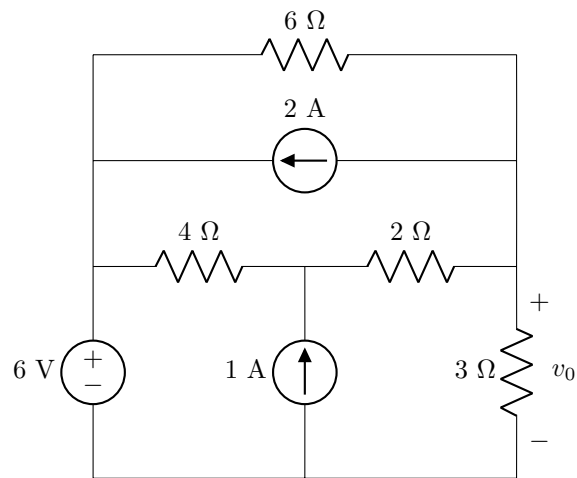
Circuit B:



01	R_x . [2]
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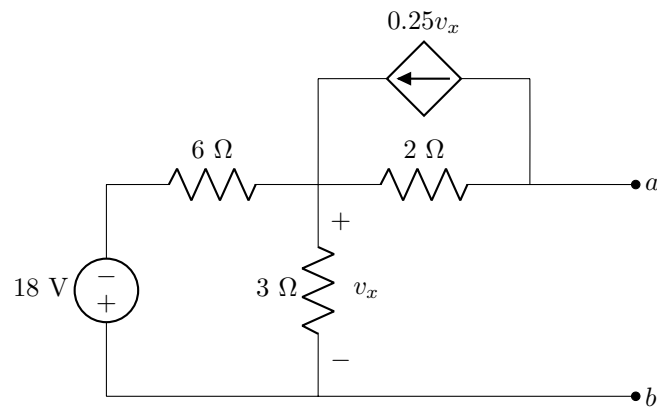
02	A . [1]
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03	B . [1]
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Questions 4 to 5 [4]Calculate the independent contributions of the 2 A and 6 V sources to the voltage v_0 .**04** Contribution to v_0 of 2 A source. [2]**05** Contribution to v_0 of 6 V source. [2]

Questions 6 to 8 [6]

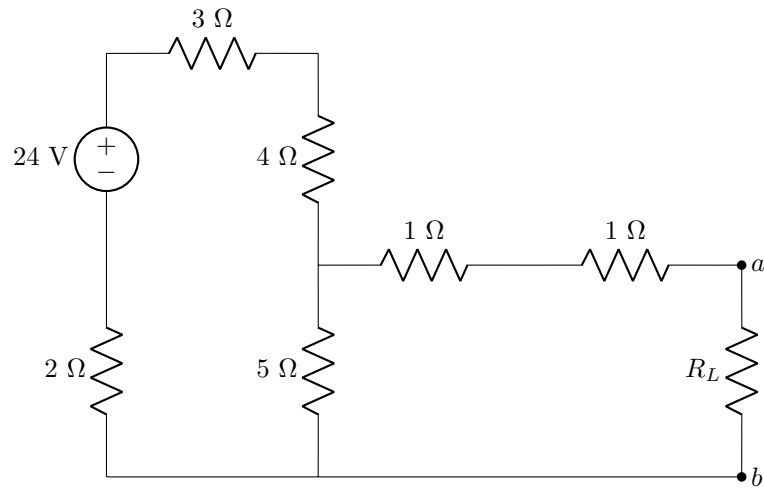
For the circuit shown, find the Norton equivalent current I_N , the Thevenin equivalent voltage V_{Th} , and the Norton equivalent resistance R_N , all with reference to terminals $a - b$.



06	I_N [2]
07	V_{Th} [2]
08	R_N [2]

Questions 9 to 10 [4]

Find the value of R_L so that maximum power is transferred to R_L ; and find p_{max} , the value of the maximum power.

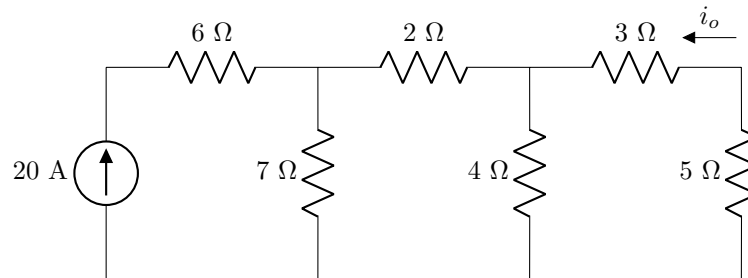


09 R_L [2]

10 p_{max} [2]

Question 11 [2]

It is known that $i_o = -4$ A for the circuit below. Find the new value of i_o if the value of the current source is increased to 27 A.

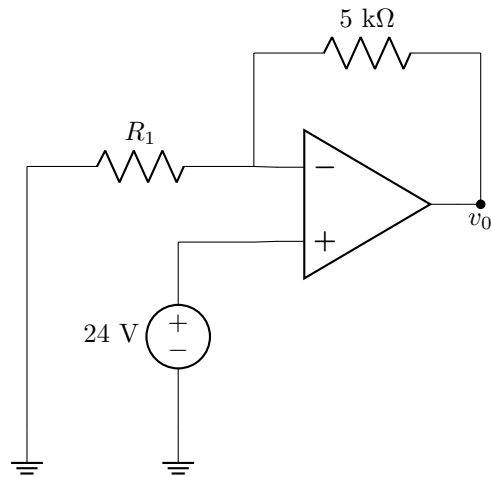


11 New value of i_o . [2]

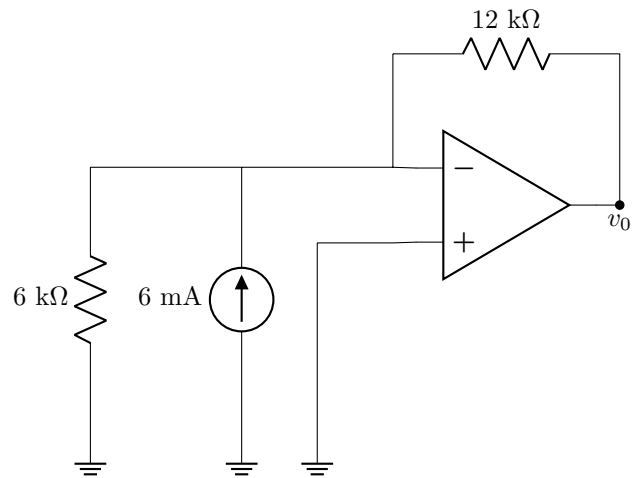
Total Section 1: [20]

Section 2**Question 12 [2]**

Find R_1 in the circuit below if $v_0 = 144$ V.

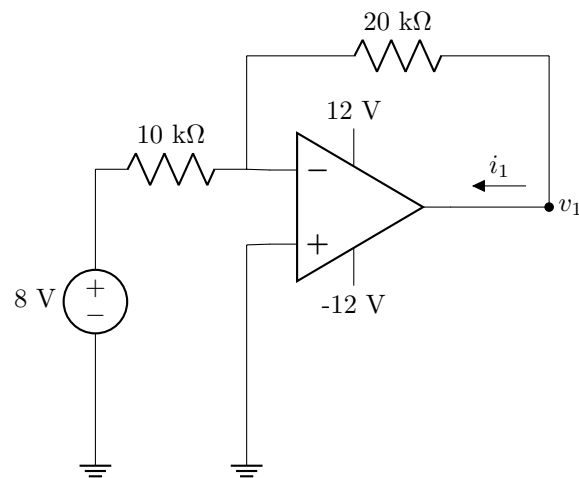


12 R_1 . [2]

Question 13 [2]Find v_0 in the circuit below.**13** v_0 . [2]

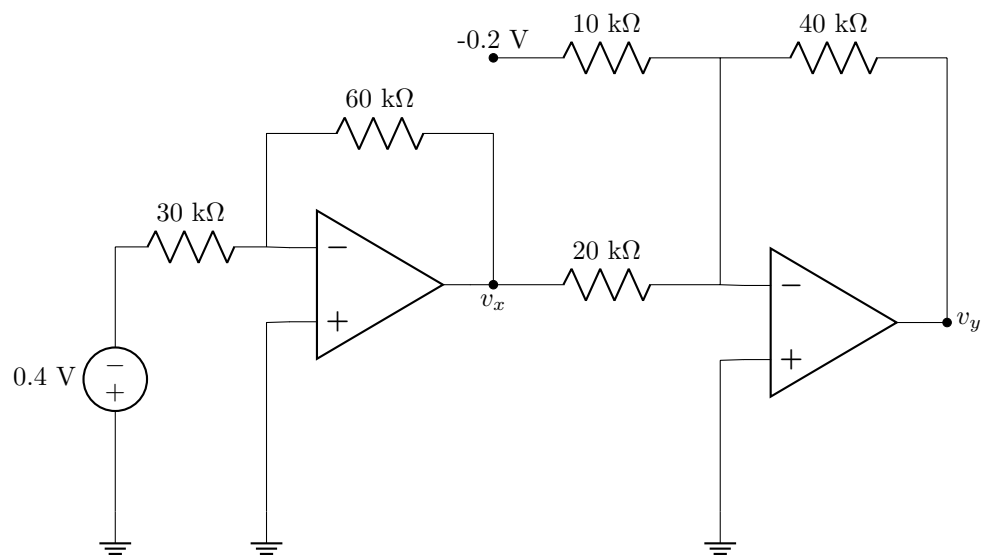
Questions 14 to 15 [4]

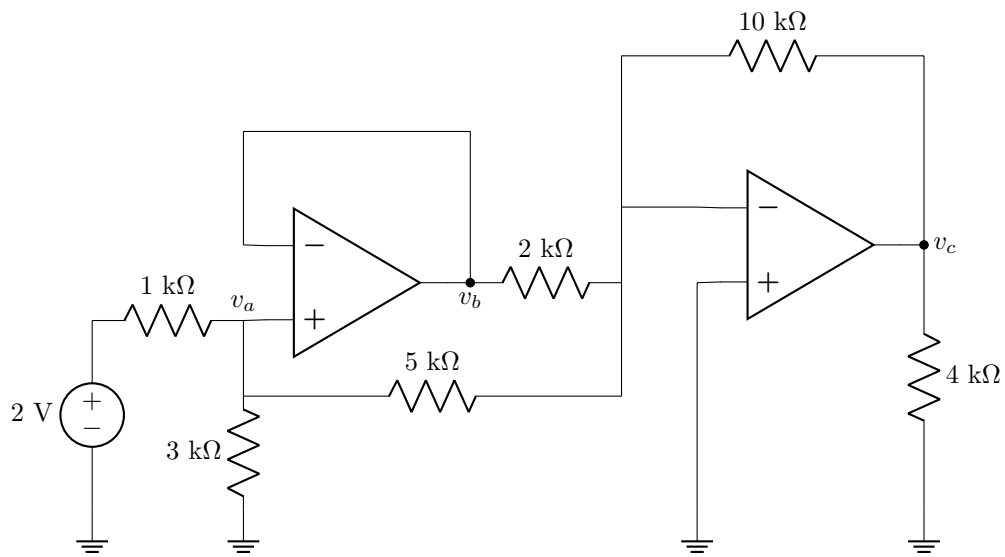
Find the voltage v_1 and the current i_1 in the circuit below. (Hint: note the op-amp supply voltages.)

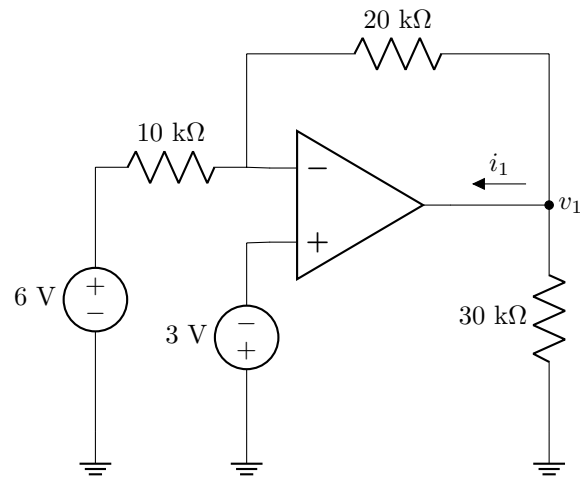


14 v_1 . [2]

15 i_1 . [2]

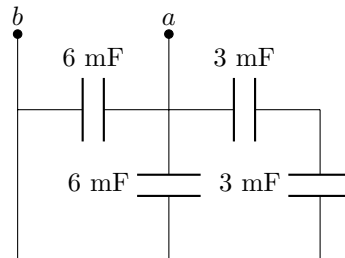
Questions 16 to 17 [3]Find the voltages v_x and v_y in the circuit below.**16** v_x . [1]**17** v_y . [2]

Questions 18 to 20 [6]Find the voltages v_a , v_b and v_c in the circuit below.**18** v_a . [2]**19** v_b . [2]**20** v_c . [2]

Questions 21 to 22 [3]Find the voltage v_1 and the current i_1 in the circuit below.**21** v_1 . [2]**22** i_1 . [1]**Total Section 2: [20]**

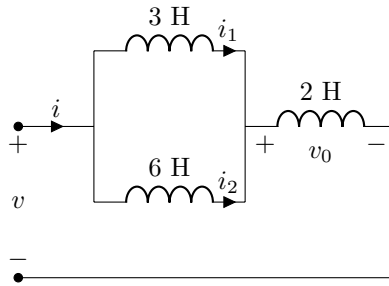
Section 3**Question 23 [2]**

Find the equivalent capacitance C_{eq} with respect to terminals $a - b$.

**23** C_{eq} . [2]

Questions 24 to 25 [4]

In the circuit below, $i_1(t) = e^{-3t}$ A and $i_2(0) = 2$ A. Determine $i_2(t = 1 \text{ sec.})$ and $v_0(t = 1 \text{ sec.})$.



24 $i_2(t = 1 \text{ sec.})$. [2]

25 $v_0(t = 1 \text{ sec.})$. [2]

This page is intentionally left open for rough work.

Total Section 3: [6]

Grand Total : [46]
